DataGuard DG-100

Multi-sensor fire detection & classification for data centres

Engineering progress briefing

Arctic Engineering · Cape Town · 22 June 2026

Prepared for investor review · Confidential

6	Built & verified	Designed	OTA tuning
sensing modalities fused on-device	full software stack (firmware → dashboard)	optical chamber ready to assemble	live re-tuning, no reflashing

DataGuard is a multi-sensor fire detection and classification platform aimed at data centres, where a fire must be caught in its earliest smouldering phase and where a false alarm that needlessly discharges suppression is itself a costly incident. Rather than trusting a single smoke sensor, DataGuard fuses six complementary sensing modalities and classifies the result on the device into a fire type and a graded confidence level before it ever raises an alarm.

This briefing summarises how the system works and the current state of engineering. The complete software platform — device firmware, cloud bridge, database and operator dashboard — is built and compile-verified, and was independently audited and security-hardened in the most recent development cycle. The custom optical sensor that differentiates the product is fully designed in mechanical model, driver software and bill of materials, and is ready for physical assembly and calibration.

01

The problem & the opportunity

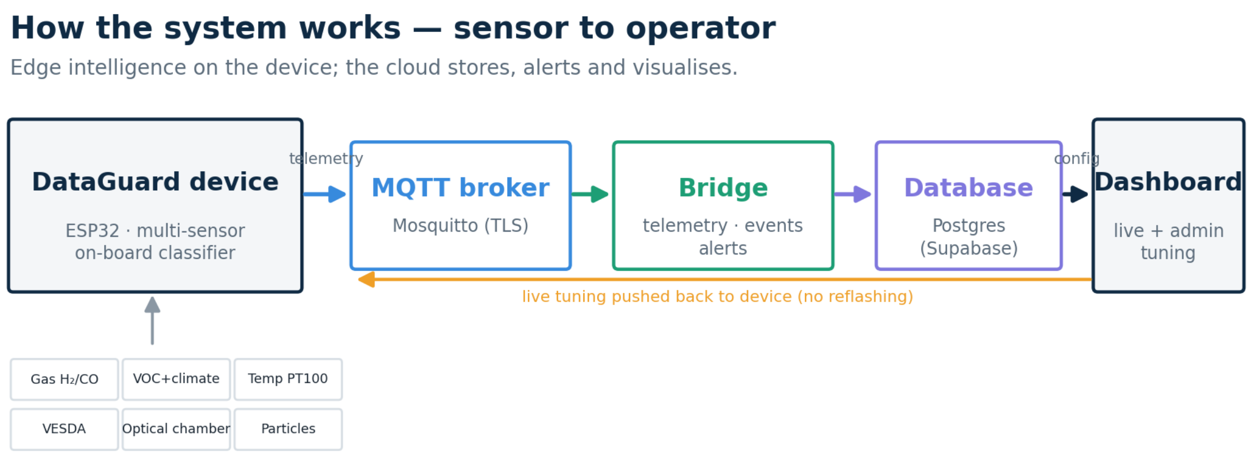
Data centres are an unusually unforgiving fire environment. High-value equipment, dense cabling and continuous uptime requirements mean detection has to happen during the earliest overheating and off-gassing stage — long before there is visible smoke. Yet a single-sensor detector that raises a false alarm and triggers gas suppression can cause an outage on its own. The result is a hard trade-off in conventional systems: react early and risk false alarms, or set thresholds high and detect late.

Conventional point smoke detectors react late. High-end aspirating detectors such as VESDA are far more sensitive but expensive and still read a single modality — they see smoke, not what is producing it. DataGuard's thesis is that combining several cheap, complementary sensors and fusing them intelligently delivers both earlier detection and far fewer false alarms than any single sensor can, at a hardware cost that scales across a large fleet.

02

How the system works

Each device carries the full sensor suite and runs the classifier locally on an ESP32 microcontroller, so a detection decision never depends on the network. Readings and classified events are published over MQTT to a cloud bridge, which stores them in a database and raises operator alerts by email and SMS. A web dashboard shows live status across a whole fleet and — importantly for commissioning and field tuning — lets an authorised operator adjust every detection parameter remotely. New settings are pushed back to the device over the air, with no need to physically reflash firmware.



Edge intelligence on the device; the cloud stores, alerts and visualises; tuning flows back over the air.

03

The sensing platform

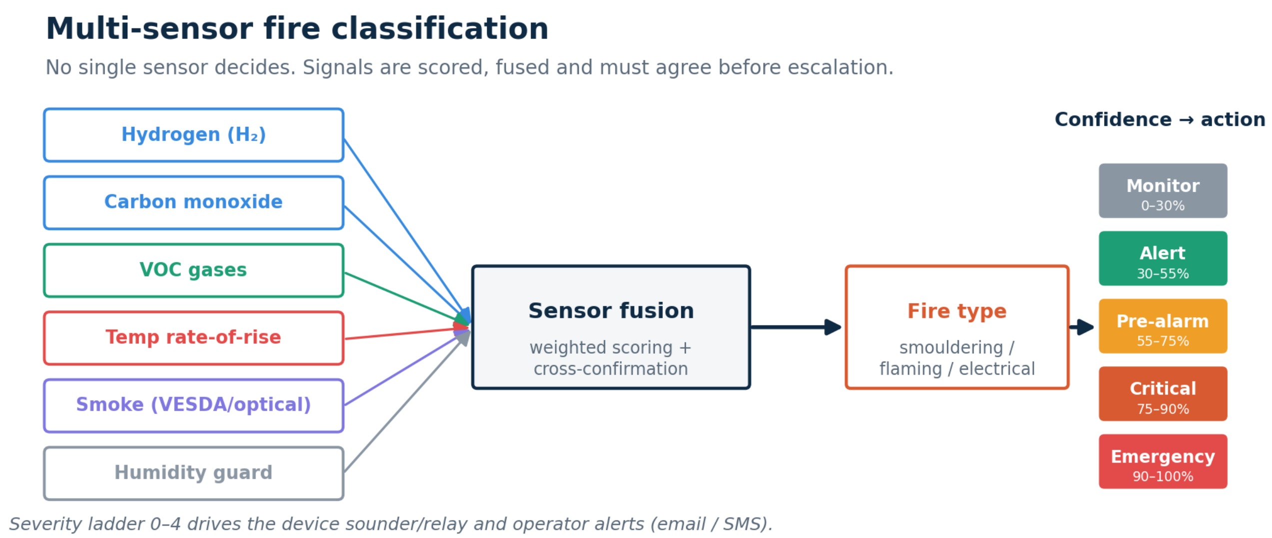
Six sensing channels give the classifier independent lines of evidence — combustion gases, air chemistry, heat, smoke and optical particle analysis.

Channel	Sensor	What it detects
Combustion gas	Alphasense H_2 + CO cells	Hydrogen & carbon monoxide from early combustion
Air chemistry	BME680	Volatile organics, temperature, humidity, pressure
Precision heat	MAX31865 + PT100	Accurate temperature & rate-of-rise
Aspirating smoke	VESDA interface	Established very-early smoke detection
Optical chamber	Custom dual-wavelength (ADPD4101)	Smoke + particle size → fire type
Particles	PMS5003 (optional)	Particle-count cross-check

04

The classification engine

The classifier never lets one sensor raise an alarm on its own. Each signal is scored, the scores are fused with cross-confirmation between sensors, and the device only escalates when enough independent evidence agrees. The fused result is mapped to a graded action ladder — monitor, alert, pre-alarm, critical, emergency — tied to a confidence percentage, while the device simultaneously estimates the fire type.



Signals are scored and fused; only agreeing evidence escalates along the confidence-to-action ladder.

The reason fusion matters is that each fire type leaves a distinct multi-sensor fingerprint. A smouldering electrical fault shows high carbon monoxide with little temperature rise; an open flame shows a sharp rate-of-rise and fine smoke particles; an overheating cable out-gasses hydrogen first. Just as important, the events that are NOT fires — cooking steam, humidity spikes — trip smoke alone but lack the gas and rate-of-rise signature, so they are rejected. This is precisely how DataGuard targets earlier detection and fewer false alarms at the same time.

Why fusion matters — each fire type has a distinct multi-sensor fingerprint

	H_2	CO	VOC	Temp rate	Smoke	Blue/IR ratio
Smouldering (early electrical)	med	high	med	low	med	low
Flaming (open fire)	low	med	high	high	high	high
Overheating cable	high	low	med	med	low	low
Cooking / steam (false alarm)	–	–	low	low	med	med

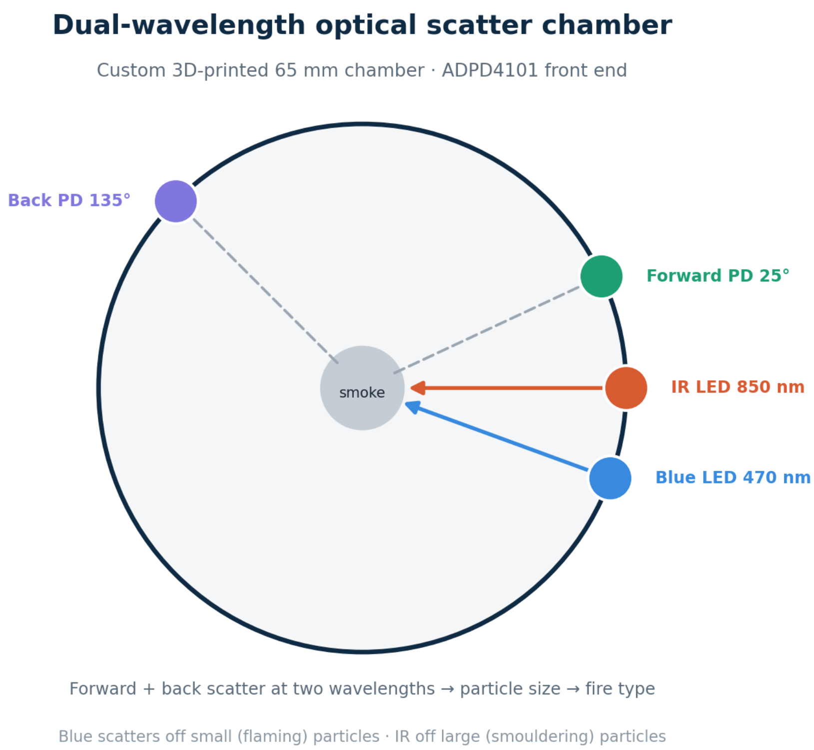
The cooking/steam row trips smoke alone but lacks the gas + rate-of-rise signature — so it is rejected, cutting false alarms.

Each fire type — and each false alarm — has a characteristic signature across the sensor set.

05

The optical innovation

The optical chamber is the part of the system custom to DataGuard. It is a 3D-printed 65 mm chamber housing two light sources — infrared and blue — and two photodiodes set at forward and backward scatter angles, all read by a single Analog Devices optical front-end. Because blue light scatters more strongly off small particles (characteristic of flaming fires) and infrared off larger particles (smouldering), measuring scatter at two wavelengths and two angles lets the device infer particle size and feed that into the fire-type decision — information a conventional smoke detector simply does not have. The optical bill of materials is modest: an evaluation front-end of roughly US$60 plus a few standard LEDs and photodiodes.



Top-down cross-section of the custom 3D-printed scatter chamber.

06

Development progress

The software foundation and the differentiating sensor design are complete. The table below reflects honest status: everything in software is built and verified; the optical chamber is fully designed and awaiting its first physical build.

Component	Status
Device firmware — sensors, classifier, runtime config	**Built · compile-ready**
Cloud bridge — MQTT to database, email/SMS alerts	**Built · verified · hardened**
Database schema & migrations	**Written · ready to apply**
Operator + admin dashboard	**Built · type-checked**
Independent security review	**Completed · fixes applied**
Optical chamber — mechanical 3D model	**Designed**
Optical chamber — driver software	**Written · awaiting bring-up**
Optical chamber — physical build	**Not started · BOM ready to order**
Field pilot	**Pending hardware**

Most recent engineering cycle

In the latest cycle the entire codebase was independently audited and the deployable issues closed: the device-command interface was secured behind authentication, message routing was hardened against injection, the cloud broker was locked to authenticated connections only, and the full telemetry data contract was reconciled end-to-end so no sensor data is lost between the device and the dashboard.

07

What comes next

1	**Assemble the first chamber.** Order the optical bill of materials and integrate the chamber onto the sensor board.
2	**Calibrate the optics.** Bench-tune the optical front-end and validate dual-wavelength particle discrimination against reference smoke.
3	**Integrate & pilot.** Connect the chamber to the live classifier and run an end-to-end test on a representative enclosure.
4	**Field trial.** Deploy in a representative data-centre environment and gather classification-performance data.

The software platform and the differentiating sensor design are done. The immediate next phase is hardware assembly, calibration and a field pilot — turning a verified design into demonstrated detection performance.